File permissions in Linux

Project description

In this project, I am serving as a security professional on a research team at a large organization. I am tasked with updating file permissions for certain files within the project's directory. I am to help keep the system secure by examining existing permissions on the file system and ensuring users on this team are authorized with the appropriate permissions.

Check file and directory details

To begin I first navigated to the projects directory:

```
cd /hom/researcher2/projects
```

Then, I entered the following command to check file and directory details:

```
ls - la
```

This command displays a detailed list of the file contents including any hidden files. This the output I received:

```
total 32
drwxr-xr-x   3 researcher2 research_team 4096 Oct       7 18:32 .
drwxr-xr-x   3 researcher2 research_team 4096 Oct       7 18:32 ..
-rw-w----    1 researcher2 research_team    46 Oct 7 18:32 .project_x.txt
drwx-x--     2 researcher2 research_team  4096 Oct 7 18:32 drafts
-rw-rw-rw-   1 researcher2 research_team        46 Oct 7 18:32 project_k.txt
-rw-r----     1 researcher2 research_team        46 Oct 7 18:32 project_m.txt
-rw-rw-r--    1 researcher2 research_team        46 Oct 7 18:32 project_r.txt
-rw-rw-r--    1 researcher2 research_team        46 Oct 7 18:32 project_t.txt
```

Describe the permissions string

Each string showcased in the output above corresponds to differing access settings. See below to determine who is authorized to access the file and their specific permissions:

- **1st character**: Either a d or hyphen (-).
    - If it's a d, it's a directory. If it's a hyphen (-), it's a regular file.

- **2nd-4th characters**: These characters correspond to the user and could either be the read (r), write (w), and execute (x).
    - If it is a hyphen (-)it indicates that this permission is not granted to the user.
- **5th-7th characters:** These characters correspond to the group and could be read (r), write (w), and/or execute (x).
    - If it is a hyphen (-) instead, then the group does not have permission.
- **8th-10th characters:** These characters correspond to others and can be either read (r), write (w), and/or execute (x).
    - If it is a hyphen (-), permission is not granted.

For example:

Each file directory can have a permission string like:
`-rw-rw-rw-`

This can be described by the following:

- 1st character: `-`
    - This means it is a file (not a directory)
- Characters 2–4: `rw-`
    - This showcases that the owner can read and write
- Characters 5–7: `rw-`
    - This demonstrates that the group can read and write
- Characters 8–10: `rw-`
    - Finally, the shows that others can read and write

Change file permissions

My company decided that other should no longer be authorized to access any of the files. To deny access to other, I utilized the following command:
`chmod o-w project_k.txt`

The command chmod changes the permissions of the files and directories. This changed `project_k.txt` from `-rw-rw-rw-rw-` to `-rw-rw-r–`.

Change file permissions on a hidden file

Then, the organization archived `.project_x.txt`. To remove write access off this project, I utilized the following linux commands:

`chmod u-w, g-w, g+r .project_x.txt`

The chmod command works to change the permissions of the file, and the commands following it removed write permissions (`g-w`) and added read permissions (`g+r`) for the group.

This changed project_x.txt from `-rw-rw-rw-` to `-r--r--r--`.

Change directory permissions

Finally, the organization also wants to have `researcher2` as the only individual with access to the `drafts` directory and its contents.

The following linux commands showcase this:
`chmod g-x drafts`

This command removes the execute permission for the group and keeps it enabled for `researcher2`. Therefore `drwx--x---` is now changed to `drwx------` drafts.

Summary

In this activity, I utilized linux commands to access file and directory permissions and alter them based on my organization's needs. By doing so I ensured authorized users had appropriate access. I used ls -la and chmod commands to check current permissions and correct them. Overall, I have now guaranteed restricted access, confidentiality, integrity, and security.